

EDWARD C. KING JR.

king8three@gmail.com | [LinkedIn](#) | [GitHub](#) | Reston, VA | U.S. Citizen

PROFESSIONAL SUMMARY

Physical security systems engineer with 20+ years designing, integrating, deploying, and commissioning enterprise access control, video surveillance (CCTV/VMS), intrusion detection, and identity-driven access systems across mission-critical environments — including global hyperscale data center campuses and major airport infrastructure. Proven record establishing standards, technical specifications, and integrated system designs, then carrying them from prototype through integration, commissioning, and hard field troubleshooting to operational delivery. Deep multi-vendor platform expertise (Honeywell ProWatch, Lenel OnGuard, CCure 9000, Milestone VMS) combined with network/infrastructure depth (TCP/IP, VLANs, AWS, Windows Server) and hands-on software-engineering ability (Java/Spring Boot, REST APIs, Python). Skilled at partnering across infrastructure, networking, cybersecurity, SOC, compliance, procurement, and vendor teams to turn ambiguous requirements into resilient, scalable security architecture at global scale.

CORE TECHNICAL COMPETENCIES

Security Architecture & Standards: Enterprise security system architecture, reference designs & technical patterns, design governance & architecture reviews, standards/specifications & design-criteria development, multi-site & global deployment models, secure deployment & lifecycle management

Security Technologies: Enterprise access control (ProWatch, Lenel OnGuard, CCure 9000), video management & CCTV/VMS analytics (Milestone), NVR appliances, intrusion detection (IDS), integrated security monitoring, identity-driven access management, PKI/PIV smart-card credentialing, ONVIF

Infrastructure & Integration: AWS environments, hyperscale & critical-infrastructure data centers, IP network design (TCP/IP, VLANs, DNS, DHCP, PoE), system hardening, multi-vendor & multi-platform integration (REST APIs, SQL), Windows Server (2012R2–2022)/Active Directory, VMware, Hyper-V, Linux/UNIX

Engineering, Automation & Docs: Java/Spring Boot, REST APIs, JWT/RBAC, Python, PowerShell, Bash, SQL/PostgreSQL, Docker, CI/CD, Git/GitHub, Agile/Scrum; control drawings, sequence of operations, functional specifications, BOM

PROFESSIONAL EXPERIENCE

Access Control Engineer | AECOM — supporting WMATA *Apr 2026 – Present*

Enterprise security-systems engineering across a multi-site transit network

- Define security-system requirements across a multi-site enterprise — establishing credential technologies, clearance codes, access architecture, and integration standards governing access to critical resources.
- Engineer and administer the Honeywell ProWatch access control platform (clearance codes, access levels, logical devices, interlocks, triggers), enforcing least-privilege, identity-driven access models across facilities.
- Architect integration of access control with CCTV/VMS, visitor management, intrusion detection, and enterprise IT — including IP network design, PoE, and VLAN segmentation across multi-vendor systems.
- Produce engineering deliverables and develop standards, specifications, and design criteria; perform QA/QC and drive defect remediation through to resolution.
- Partner with contractors, IT teams, and client leadership through design and implementation, presenting technical solutions and driving cross-team architecture decisions.

Security Program Manager | Amazon Web Services (AWS) *May 2025 – Apr 2026*

Physical security architecture across global hyperscale data center programs

- Led security-systems engineering and program delivery for enterprise access control, VMS, and IDS architecture across regional data center new-construction, expansion, and retrofit programs — spanning diverse deployment models at hyperscale.
- Evaluated emerging physical security technologies (CCTV, access control, intrusion detection), assessed alternative architectures, and provided SME design governance on integrated platform decisions aligned to infrastructure risk profile.
- Established deployment, resilience, and lifecycle-management standards; produced bid packages, technical specifications, control drawings, and code-compliance verification across the project lifecycle.
- Managed security-product vendors and integrators through the full lifecycle (on-boarding, RFQ, bid evaluation, negotiation, QA/QC, support) with multi-million-dollar budget authority, coordinating cross-functional IT, facilities, networking, and security teams.

Senior Systems Engineer | Pavion (embedded — Philadelphia Intl. Airport) *Nov 2023 – May 2025*

Tier 2/3 engineering for integrated security infrastructure serving 1,000+ users

- Delivered Tier 2/3 engineering for integrated CCTV/VMS, access control, intrusion detection, and Windows Server infrastructure, sustaining 99.9% uptime in a mission-critical airport environment.
- Built PowerShell and Python automation for infrastructure monitoring, health checks, and incident detection — reducing response time by 40%.
- Implemented proactive integrated monitoring with automated alerting via REST APIs and SQL integration for security event correlation across multi-platform systems.
- Resolved complex integration issues through systematic log analysis with IT infrastructure, facilities, and contractor teams.

Senior Security Systems Administrator | Advanced Data & Risk Mgmt (UPenn) *Feb 2023 – Nov 2023*

- Administered integrated CCTV/VMS, access control, and IDS in complex Windows Server / VMware environments supporting 1,000+ users at 99.9% uptime.
- Performed Active Directory administration, security-policy enforcement, and authentication troubleshooting; implemented automated procedures to validate system integrity.

Enterprise Technician V | ADT Commercial *Mar 2022 – Feb 2023*

- Designed and implemented enterprise CCTV, access control, and IDS integrations for commercial and multi-site clients; built SQL/PowerShell reporting and monitoring tools and delivered stakeholder presentations.

Installation Technician IV | Johnson Controls *Oct 2019 – Mar 2022*

- Designed and installed enterprise CCTV, access control, and IDS for commercial/industrial facilities; reviewed construction documents, led programming, commissioning, and testing; standardized PowerShell deployment scripts for consistent configuration at scale.

Service Technician | Siemens Building Technologies *Apr 2018 – Oct 2019*

- Installed, programmed, and commissioned IP cameras, access control panels, and integrated security devices per engineering specifications across commercial and industrial environments.

Installation / Commissioning Technician | Security and Data Technologies *Aug 2015 – Apr 2018*

- Led installation, integration, programming, and commissioning of enterprise CCTV, access control, and IDS; supervised contractor crews to engineering drawings and performed final commissioning, testing, and client handoff.

Earlier roles: Security Instrument (2014–2015) | Guardian Protection Services (2013–2014) | BSAFE Alarms (2010–2013) | Ingersoll Rand (2010) | Security and Data Technologies — Fire Alarm Tester/Inspector (2008–2009). Full hands-on foundation across CCTV, access control, IDS, and fire-alarm commissioning.

SOFTWARE ENGINEERING PROJECTS

- **PhySec.Dash**— Full-stack security operations dashboard. I utilized Java/Spring Boot REST API with JWT authentication and role-based access control (RBAC), React front end, deployed via CI/CD to Railway and Vercel with a live demo. Demonstrates n-tier architecture, secure auth, and end-to-end lifecycle ownership. Source: <https://physec-dashboard.vercel.app/>
- **PhySec.Learn** — I designed this with the goal to be a learning platform for physical access control systems. Utilizing Next.js (TypeScript) for the fullstack and Prisma/PostgreSql for database management. Will be deployed on Vercel in the coming days. Source: <https://open-accs.vercel.app/>

EDUCATION & CERTIFICATIONS

Certifications: Honeywell ProWatch · Lenel OnGuard · CCure 9000 · Milestone VMS · Google Cybersecurity Certificate (2023)

Education: Full-Stack Software Development — LaunchCode (2025); REST APIs, SQL integration, system architecture

Additional: NIST Cybersecurity Framework · Splunk SIEM · U.S. Citizen · Eligible for Secret clearance